Python Basics Questions

1. What is Python, and why is it popular?

Python is a high-level, easy-to-learn programming language.
It's popular because it has:

- Simple syntax (like English)
- Huge libraries
- Wide use (web, data science, automation, AI)
- Great community support

2. What is an interpreter in Python?

An **interpreter** is a program that reads and runs Python code **line by line**, helping you test and debug easily.

3. What are pre-defined keywords in Python?

Keywords are special reserved words in Python (like if, for, def) that have specific meaning.
You cannot use them as variable names.

4. Can keywords be used as variable names?

No, keywords **cannot** be used as variable names — it will cause a syntax error.

5. What is mutability in Python?

Mutability means whether the value of an object **can be changed** after it's created.

- **Mutable**: can change (e.g. list)
- **Immutable**: cannot change (e.g. tuple, str)

6. Why are lists mutable but tuples are immutable?

- **Lists** are designed to hold changing data — you can add, remove, or update items.
- **Tuples** are used for fixed data — safer and faster, especially as dictionary keys.

7. What is the difference between == and is?

- == checks if values are the **same**
- is checks if both refer to the **same object in memory**

8. What are logical operators in Python?

Used for combining conditions:

- and → True if both are True
- or → True if at least one is True
- not → reverses the result

9. What is type casting in Python?

Type casting means converting data from one type to another (e.g. str to int).

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```
int("5")  # changes string to integer 5
```

10. What is the difference between implicit and explicit type casting?

- **Implicit**: Python converts automatically
- **Explicit**: You convert it using functions like int(), str(), etc.

11. What is the purpose of conditional statements in Python?

Conditional statements like if, elif, else help your program **make decisions** based on conditions.

12. How does the elif statement work?

elif (else-if) is used to test **multiple conditions** one after the other. Python checks top to bottom and runs the first condition that's true.

13. What is the difference between for and while loops?

- for loop → used when you **know how many times** to loop
- while loop → used when you **loop until a condition** is false

14. Scenario where while loop is better than for loop?

Example: keep asking for password until it’s correct.

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```
password = ""
while password != "1234":
    password = input("Enter password: ")
```

You don’t know how many times user will try — that’s why while is better.

Practical Questions

1. Print "Hello, World!"

```
print("Hello, World!")
```

2. Display your name and age

```
name = "Vishal"
age = 25
print("My name is", name)
print("My age is", age)
```

3. Print all pre-defined keywords

```
import keyword
print("Python Keywords are:")
print(keyword.kwlist)
```

4. Check if a word is a Python keyword

```
import keyword
word = input("Enter a word: ")
if keyword.iskeyword(word):
    print(f"{word} is a Python keyword.")
else:
    print(f"{word} is NOT a Python keyword.")
```

5. List vs Tuple (mutability demo)

```
# List (mutable)
my_list = [10, 20, 30]
my_list[1] = 99  # This works
print("Modified List:", my_list)

# Tuple (immutable)
my_tuple = (10, 20, 30)
# my_tuple[1] = 99  # ❌ Error: Tuples can't be changed
print("Tuple:", my_tuple)
```

6. Mutable vs Immutable Function Arguments

```
def change_list(lst):
    lst[0] = "changed"

def change_number(n):
    n = 100
my_list = [1, 2, 3]
my_number = 10
change_list(my_list)
change_number(my_number)
print("List after function:", my_list)     # Changed
print("Number after function:", my_number) # Unchanged
```

7. Basic Arithmetic Operations

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
print("Sum:", a + b)
print("Difference:", a - b)
print("Product:", a * b)
print("Quotient:", a / b)
```

8. Use of Logical Operators

```
x = 10
y = 20

print(x > 5 and y > 15)   # True
print(x > 15 or y > 15)   # True
print(not(x > 15))       # True
```

9. Convert user input to int, float, bool

```
user_input = input("Enter something: ")
int_val = int(user_input)
float_val = float(user_input)
bool_val = bool(user_input)

print("Integer:", int_val)
print("Float:", float_val)
print("Boolean:", bool_val)
```

10. Type Casting with List Elements

```
str_list = ["1", "2", "3"]

int_list = [int(i) for i in str_list]

print("String List:", str_list)

print("Converted to Integers:", int_list)
```

11. Check if a number is positive, negative, or zero

```
num = float(input("Enter a number: "))


if num > 0:

   print("Positive")

elif num < 0:

   print("Negative")

else:

   print("Zero")
```

12. For loop to print 1 to 10

```
for i in range(1, 11):

   print(i)
```

13. Sum of all even numbers from 1 to 50

```
total = 0
for i in range(1, 51):
    if i % 2 == 0:
        total += i
print("Sum of even numbers:", total)
```

14. Reverse a string using while loop

```
text = input("Enter a string: ")
reversed_text = ""
i = len(text) - 1
while i >= 0:
    reversed_text += text[i]
    i -= 1
print("Reversed String:", reversed_text)
```

15. Factorial using while loop

```
num = int(input("Enter a number: "))
fact = 1
i = 1
while i <= num:
    fact *= i
    i += 1
print("Factorial:", fact)
```